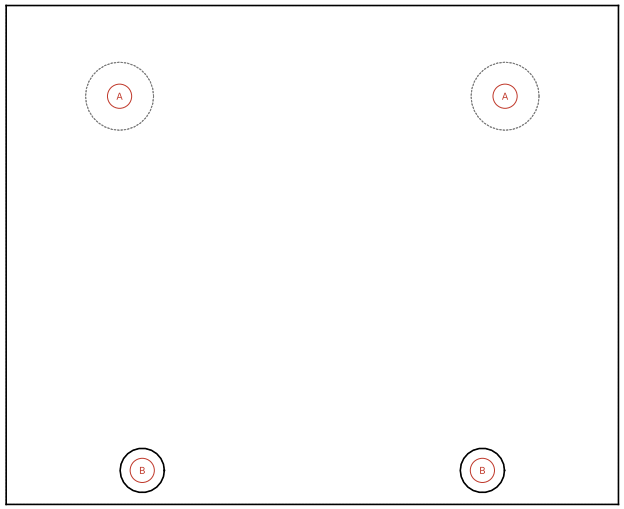
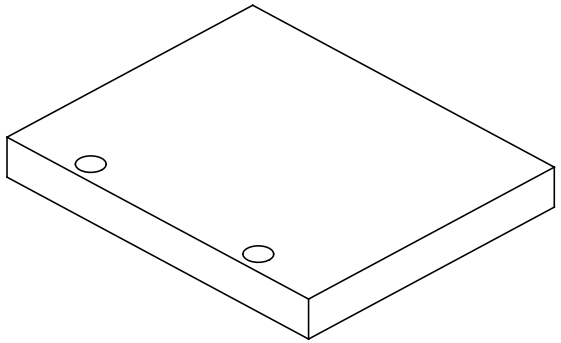
CA-203 Top cap

NOTES

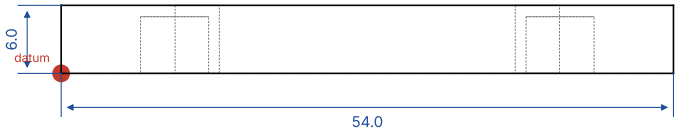
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

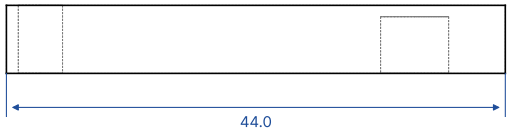




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	6.00	2	Z	5.0	(10.0, 36.0); (44.0, 36.0)	6 mm rod press fit
B	3.90	2	Z	6.0	(12.0, 3.0); (42.0, 3.0)	#6-32 clearance / running

PART NO.	**CA-203**
TITLE	**Top cap**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	10.5 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	3:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/top_cap.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	6 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02